

Initiating Search

November 4, 2025, 11:47 AM

• Search:

CAS Lexicon Search:

Decarboxylative arylation - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (59)	 References	View Results
Exported: Retrieved Related Reaction Results (161)	 Reactions	View Results

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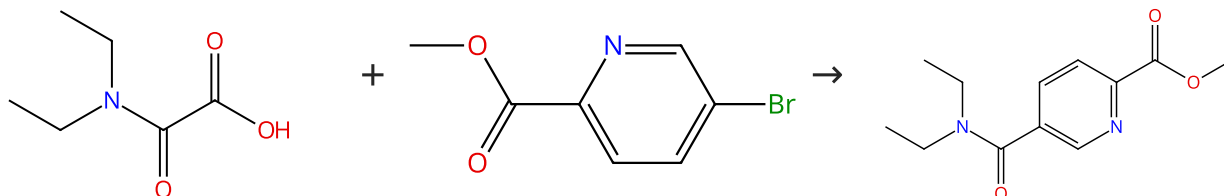
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (46)

Suppliers (102)

31-614-CAS-38656524

Steps: 1 Yield: 94%

Photoredox Ni-catalyzed decarboxylative arylation of oxamic acids for synthesis

By: Duan, Depeng; et al

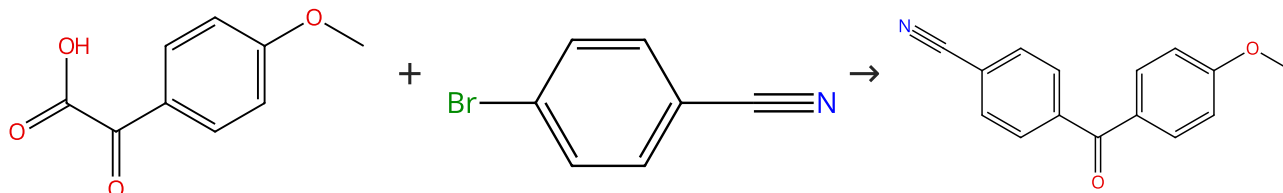
Organic Chemistry Frontiers (2024), 11(1), 47-52.

- 1.1 **Catalysts:** 4,4'-Dimethoxy-2,2'-bipyridine, Dibromo[1,2-di(methoxy-κO)ethane]nickel
Solvents: Acetonitrile; 30 min, rt
- 1.2 **Reagents:** Sodium carbonate
Catalysts: Benzonitrile, 2,3,4,5,6-penta-9*H*-carbazol-9-yl-
Solvents: 1,4-Dioxane; 12 h, rt

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (55)

Suppliers (92)

Suppliers (41)

31-484-CAS-22486057

Steps: 1 Yield: 93%

Acyl Radicals from α-Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones

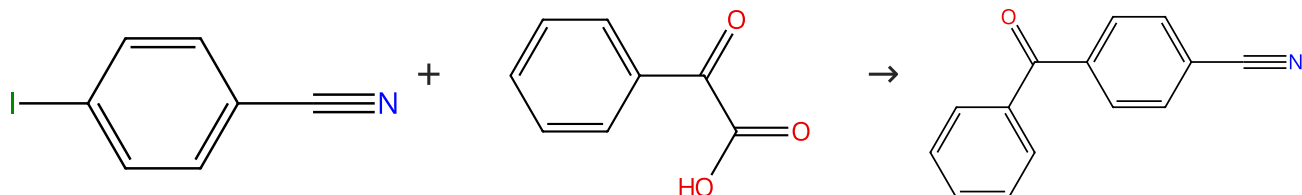
By: Zhu, Da-Liang; et al

Organic Letters (2020), 22(17), 6832-6837.

- 1.1 **Reagents:** Lithium carbonate (Li₂CO₃)
Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Dimethylformamide, Water; 24 h, cooled

Scheme 3 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (80)

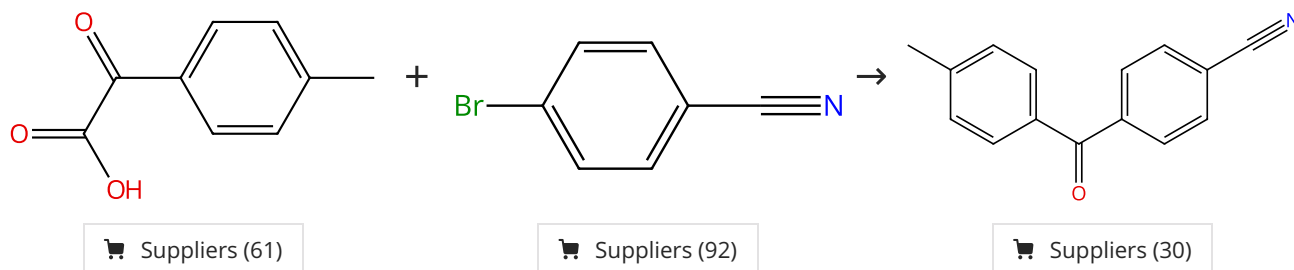
Suppliers (105)

Suppliers (63)

31-484-CAS-22486030 Steps: 1 Yield: 93% 1.1 Reagents: Lithium carbonate (Li_2CO_3) Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Dimethylformamide, Water; 24 h, cooled	Acyl Radicals from α-Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones By: Zhu, Da-Liang; et al Organic Letters (2020), 22(17), 6832-6837.
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Scheme 4 (1 Reaction)

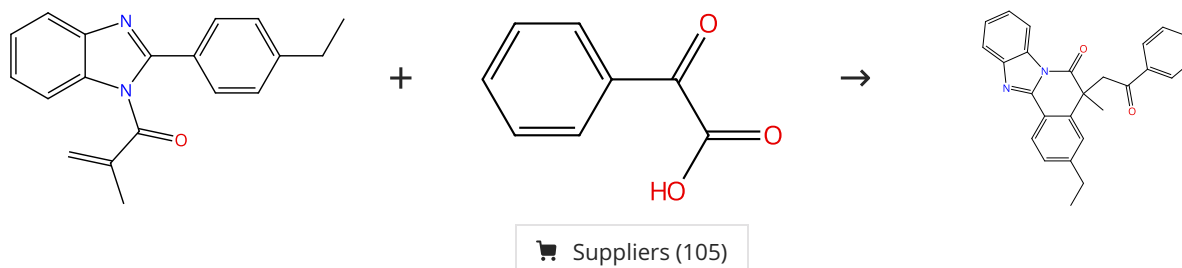
Steps: 1 Yield: 92%



31-484-CAS-22486056 Steps: 1 Yield: 92% 1.1 Reagents: Lithium carbonate (Li_2CO_3) Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Dimethylformamide, Water; 24 h, cooled	Acyl Radicals from α-Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones By: Zhu, Da-Liang; et al Organic Letters (2020), 22(17), 6832-6837.
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Scheme 5 (1 Reaction)

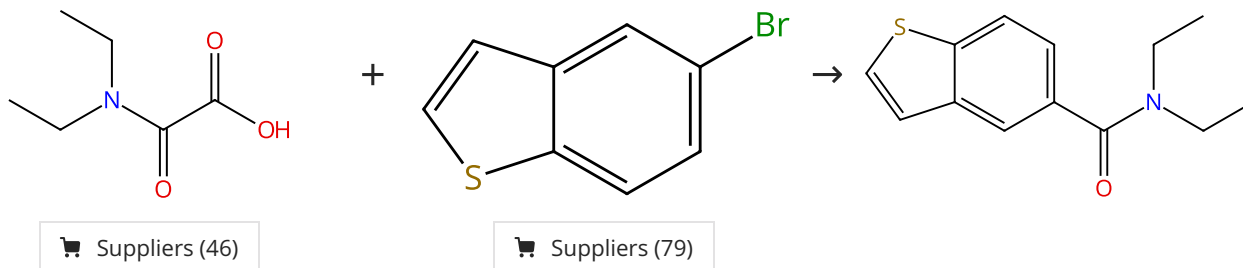
Steps: 1 Yield: 92%



31-484-CAS-23433962 Steps: 1 Yield: 92% 1.1 Reagents: Oxygen Catalysts: 2,4,5,6-Tetra(9H-carbazol-9-yl)isophthalonitrile Solvents: Dimethyl carbonate; 6 h, rt Experimental Protocols	Acyl Radicals from α-Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles By: Zhu, Hu-Lin; et al Organic Letters (2021), 23(8), 2976-2980.
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Scheme 6 (1 Reaction)

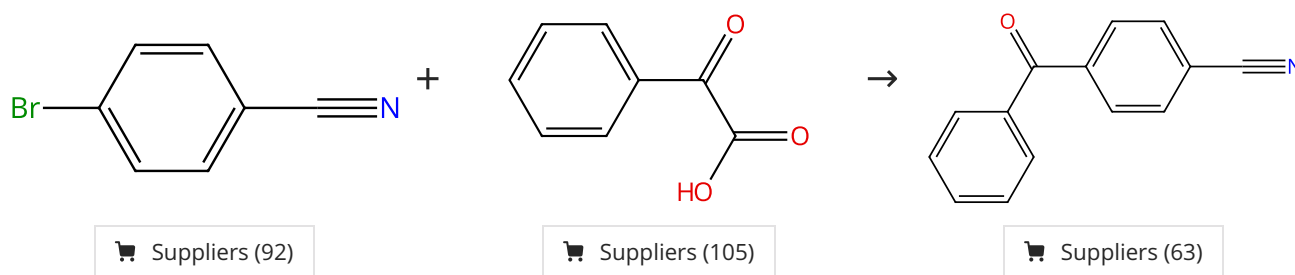
Steps: 1 Yield: 91%



31-614-CAS-38656532 Steps: 1 Yield: 91% 1.1 Catalysts: 4,4'-Dimethoxy-2,2'-bipyridine, Dibromo[1,2-di(methoxy-κO)ethane]nickel Solvents: Acetonitrile; 30 min, rt 1.2 Reagents: Sodium carbonate Catalysts: Benzonitrile, 2,3,4,5,6-penta-9<i>H</i>-carbazol-9-yl- Solvents: 1,4-Dioxane; 12 h, rt Experimental Protocols	Photoredox Ni-catalyzed decarboxylative arylation of oxamic acids for synthesis By: Duan, Depeng; et al Organic Chemistry Frontiers (2024), 11(1), 47-52.
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Scheme 7 (1 Reaction)

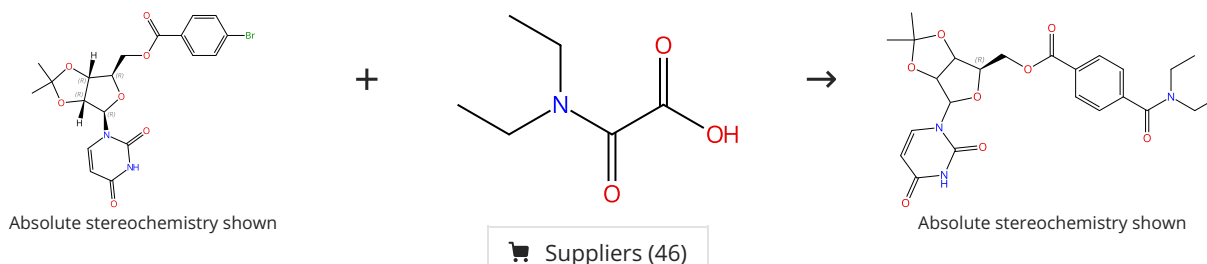
Steps: 1 Yield: 91%



31-484-CAS-22486029 Steps: 1 Yield: 91% 1.1 Reagents: Lithium carbonate (Li₂CO₃) Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Dimethylformamide, Water; 24 h, cooled	Acyl Radicals from α-Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones By: Zhu, Da-Liang; et al Organic Letters (2020), 22(17), 6832-6837.
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Scheme 8 (1 Reaction)

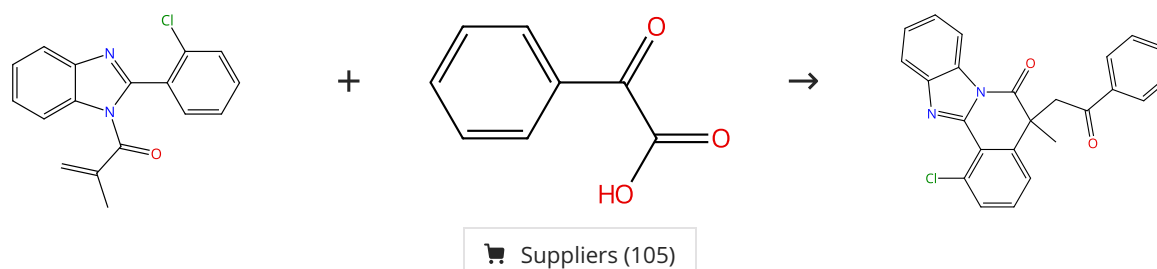
Steps: 1 Yield: 90%



31-614-CAS-38656539 Steps: 1 Yield: 90% 1.1 Catalysts: 4,4'-Dimethoxy-2,2'-bipyridine, Dibromo[1,2-di(methoxy-κO)ethane]nickel Solvents: Acetonitrile; 30 min, rt 1.2 Reagents: Sodium carbonate Catalysts: Benzonitrile, 2,3,4,5,6-penta-9<i>H</i>-carbazol-9-yl- Solvents: 1,4-Dioxane; 12 h, rt Experimental Protocols	Photoredox Ni-catalyzed decarboxylative arylation of oxamic acids for synthesis By: Duan, Depeng; et al Organic Chemistry Frontiers (2024), 11(1), 47-52.
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Scheme 9 (1 Reaction)

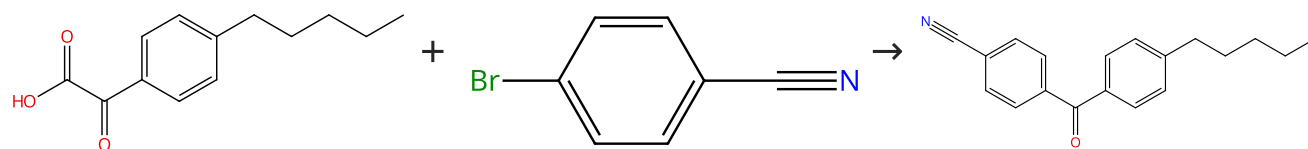
Steps: 1 Yield: 90%



31-484-CAS-23434582 Steps: 1 Yield: 90%	Acyl Radicals from α-Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles By: Zhu, Hu-Lin; et al Organic Letters (2021), 23(8), 2976-2980.
1.1 Reagents: Oxygen Catalysts: 2,4,5,6-Tetra(9 <i>H</i> -carbazol-9-yl)isophthalonitrile Solvents: Dimethyl carbonate; 6 h, rt	

Scheme 10 (1 Reaction)

Steps: 1 Yield: 90%

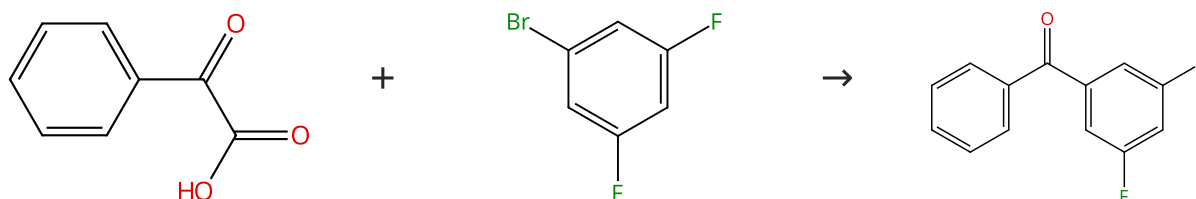


Suppliers (92)

31-484-CAS-22486058 Steps: 1 Yield: 90%	Acyl Radicals from α-Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones By: Zhu, Da-Liang; et al Organic Letters (2020), 22(17), 6832-6837.
1.1 Reagents: Lithium carbonate (Li_2CO_3) Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy- κ O)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Dimethylformamide, Water; 24 h, cooled	

Scheme 11 (1 Reaction)

Steps: 1 Yield: 89%



Suppliers (105)

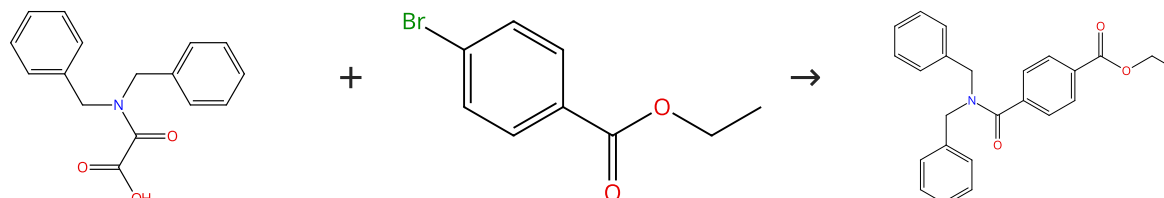
Suppliers (79)

Suppliers (51)

31-484-CAS-22486042 Steps: 1 Yield: 89%	Acyl Radicals from α-Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones By: Zhu, Da-Liang; et al Organic Letters (2020), 22(17), 6832-6837.
1.1 Reagents: Lithium carbonate (Li_2CO_3) Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy- κ O)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Dimethylformamide, Water; 24 h, cooled	

Scheme 12 (1 Reaction)

Steps: 1 Yield: 88%



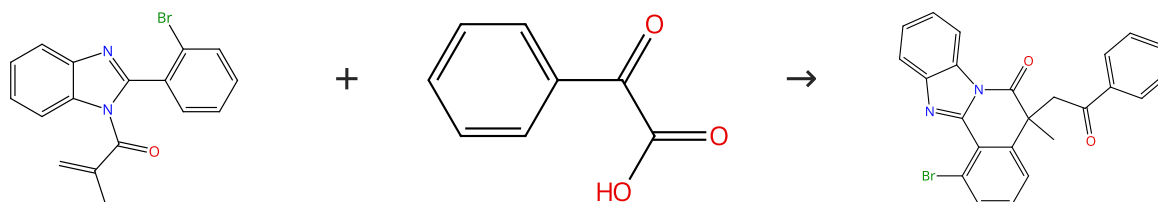
Suppliers (3)

Suppliers (81)

31-614-CAS-38656538 Steps: 1 Yield: 88% 1.1 Catalysts: 4,4'-Dimethoxy-2,2'-bipyridine, Dibromo[1,2-di(methoxy-κO)ethane]nickel Solvents: Acetonitrile; 30 min, rt 1.2 Reagents: Sodium carbonate Catalysts: Benzonitrile, 2,3,4,5,6-penta-9<i>H</i>-carbazol-9-yl- Solvents: 1,4-Dioxane; 12 h, rt Experimental Protocols	Photoredox Ni-catalyzed decarboxylative arylation of oxamic acids for synthesis By: Duan, Depeng; et al Organic Chemistry Frontiers (2024), 11(1), 47-52.
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Scheme 13 (1 Reaction)

Steps: 1 Yield: 88%

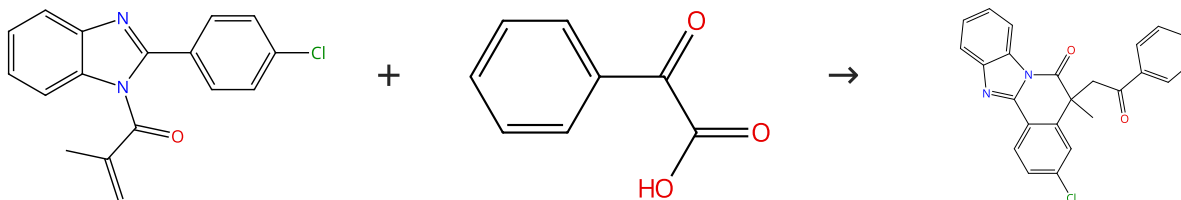


Suppliers (105)

31-484-CAS-23432841 Steps: 1 Yield: 88% 1.1 Reagents: Oxygen Catalysts: 2,4,5,6-Tetra(9<i>H</i>-carbazol-9-yl)isophthalonitrile Solvents: Dimethyl carbonate; 6 h, rt	Acyl Radicals from α-Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles By: Zhu, Hu-Lin; et al Organic Letters (2021), 23(8), 2976-2980.
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Scheme 14 (1 Reaction)

Steps: 1 Yield: 88%

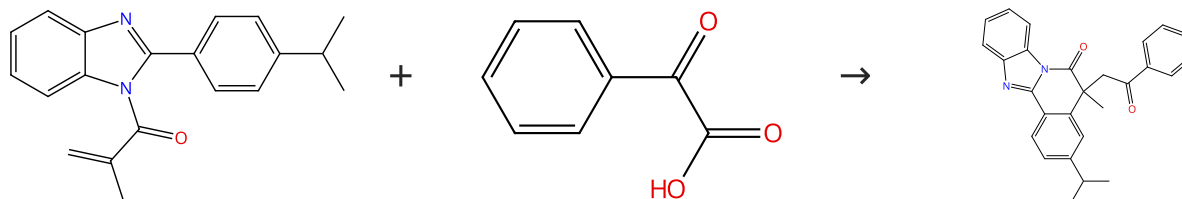


Suppliers (105)

31-484-CAS-23432601 Steps: 1 Yield: 88% 1.1 Reagents: Oxygen Catalysts: 2,4,5,6-Tetra(9<i>H</i>-carbazol-9-yl)isophthalonitrile Solvents: Dimethyl carbonate; 6 h, rt	Acyl Radicals from α-Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles By: Zhu, Hu-Lin; et al Organic Letters (2021), 23(8), 2976-2980.
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Scheme 15 (1 Reaction)

Steps: 1 Yield: 87%

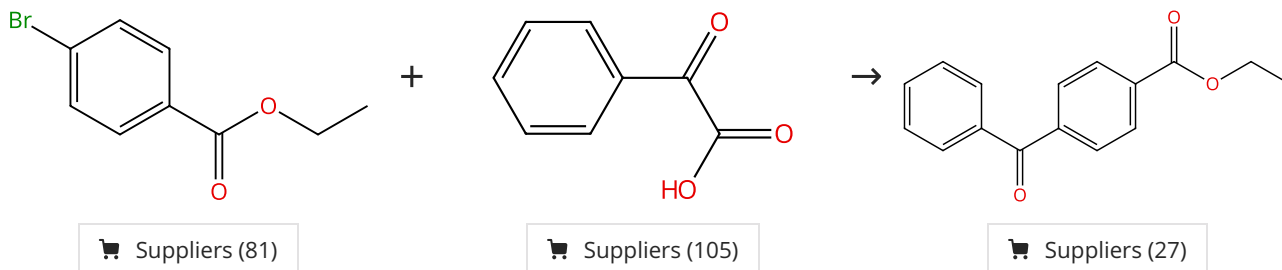


Suppliers (105)

31-484-CAS-23433919	Steps: 1 Yield: 87%	Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles
1.1 Reagents: Oxygen Catalysts: 2,4,5,6-Tetra(9 <i>H</i> -carbazol-9-yl)isophthalonitrile Solvents: Dimethyl carbonate; 6 h, rt		By: Zhu, Hu-Lin; et al Organic Letters (2021), 23(8), 2976-2980.
Experimental Protocols		

Scheme 16 (1 Reaction)

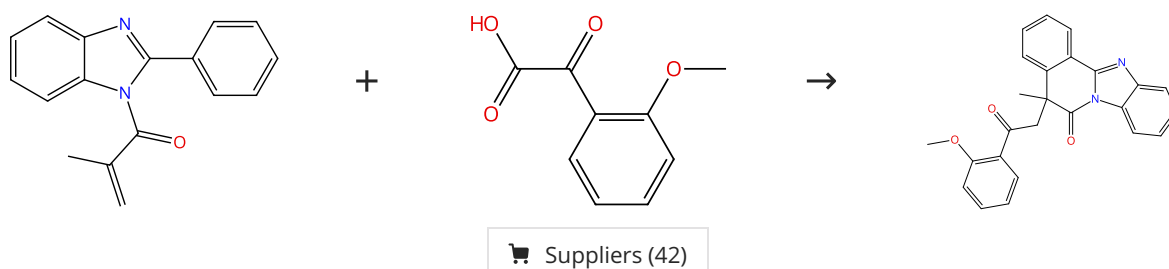
Steps: 1 Yield: 87%



31-484-CAS-22486034	Steps: 1 Yield: 87%	Acyl Radicals from α -Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones
1.1 Reagents: Lithium carbonate (Li_2CO_3) Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy- κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Dimethylformamide, Water; 24 h, cooled		By: Zhu, Da-Liang; et al Organic Letters (2020), 22(17), 6832-6837.

Scheme 17 (1 Reaction)

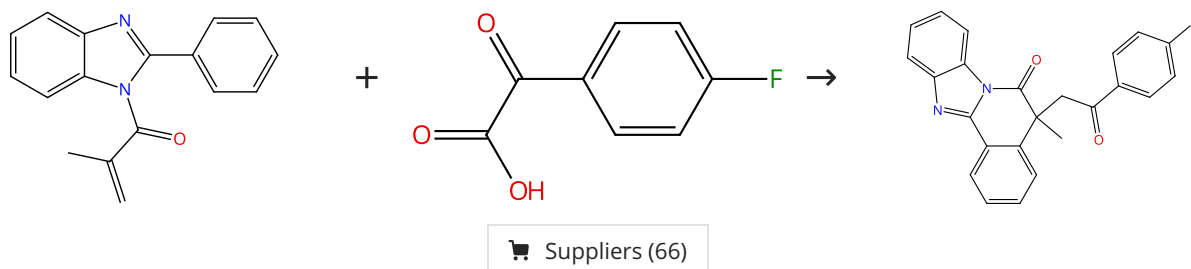
Steps: 1 Yield: 87%



31-484-CAS-23433556	Steps: 1 Yield: 87%	Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles
1.1 Reagents: Oxygen Catalysts: 2,4,5,6-Tetra(9 <i>H</i> -carbazol-9-yl)isophthalonitrile Solvents: Dimethyl carbonate; 6 h, rt		By: Zhu, Hu-Lin; et al Organic Letters (2021), 23(8), 2976-2980.
Experimental Protocols		

Scheme 18 (1 Reaction)

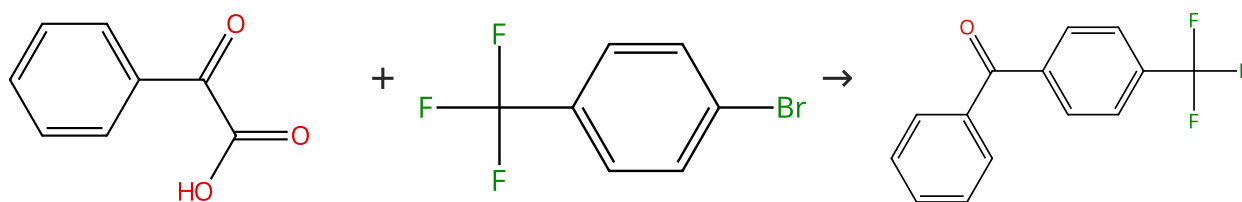
Steps: 1 Yield: 87%



31-484-CAS-23433074	Steps: 1 Yield: 87%	Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles
1.1 Reagents: Oxygen Catalysts: 2,4,5,6-Tetra(9 <i>H</i> -carbazol-9-yl)isophthalonitrile Solvents: Dimethyl carbonate; 6 h, rt		By: Zhu, Hu-Lin; et al Organic Letters (2021), 23(8), 2976-2980.
Experimental Protocols		

Scheme 19 (1 Reaction)

Steps: 1 Yield: 86%



Suppliers (105)

Suppliers (78)

Suppliers (58)

31-484-CAS-22486031

Steps: 1 Yield: 86%

1.1 **Reagents:** Lithium carbonate (Li_2CO_3)
Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy- κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Dimethylformamide, Water; 24 h, cooled

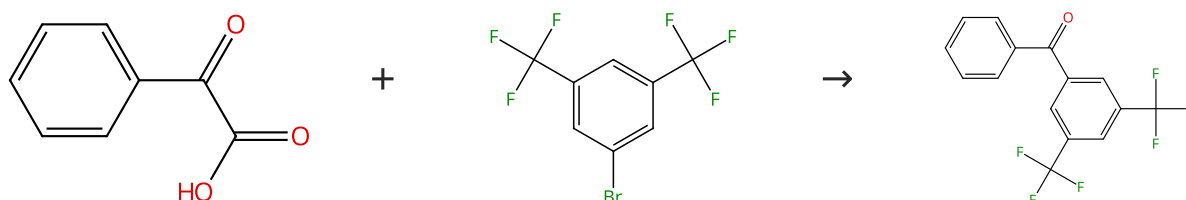
Acyl Radicals from α -Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones

By: Zhu, Da-Liang; et al

Organic Letters (2020), 22(17), 6832-6837.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 86%



Suppliers (105)

Suppliers (79)

Suppliers (48)

31-484-CAS-22486040

Steps: 1 Yield: 86%

1.1 **Reagents:** Lithium carbonate (Li_2CO_3)
Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy- κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Dimethylformamide, Water; 24 h, cooled

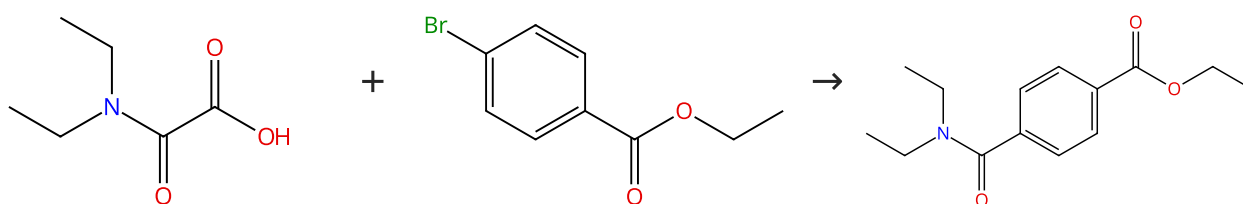
Acyl Radicals from α -Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones

By: Zhu, Da-Liang; et al

Organic Letters (2020), 22(17), 6832-6837.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 86%



Suppliers (46)

Suppliers (81)

Supplier (1)

31-614-CAS-38656508

Steps: 1 Yield: 86%

1.1 **Catalysts:** 4,4'-Dimethoxy-2,2'-bipyridine, Dibromo[1,2-di(methoxy- κO)ethane]nickel
Solvents: Acetonitrile; 30 min, rt
 1.2 **Reagents:** Sodium carbonate
Catalysts: Benzonitrile, 2,3,4,5,6-penta-9*H*-carbazol-9-yl-
Solvents: 1,4-Dioxane; 12 h, rt

Photoredox Ni-catalyzed decarboxylative arylation of oxamic acids for synthesis

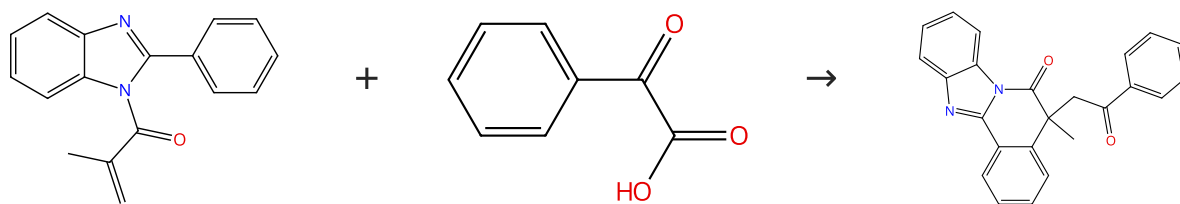
By: Duan, Depeng; et al

Organic Chemistry Frontiers (2024), 11(1), 47-52.

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 86%



Suppliers (105)

31-484-CAS-23433395

Steps: 1 Yield: 86%

Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles

By: Zhu, Hu-Lin; et al

Organic Letters (2021), 23(8), 2976-2980.

1.1 Reagents: Oxygen

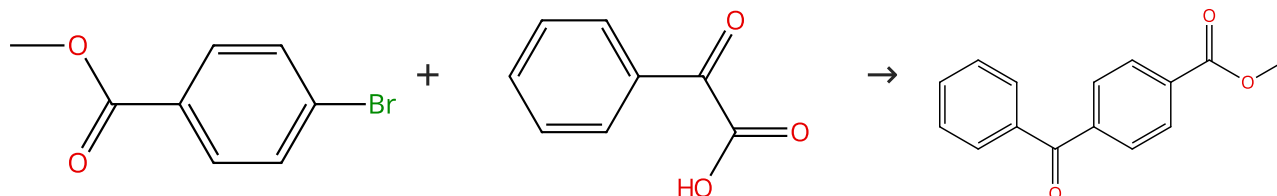
Catalysts: 2,4,5,6-Tetra(9H-carbazol-9-yl)isophthalonitrile

Solvents: Dimethyl carbonate; 6 h, rt

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 85%



Suppliers (97)

Suppliers (105)

Suppliers (42)

31-484-CAS-22486033

Steps: 1 Yield: 85%

Acyl Radicals from α -Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones

By: Zhu, Da-Liang; et al

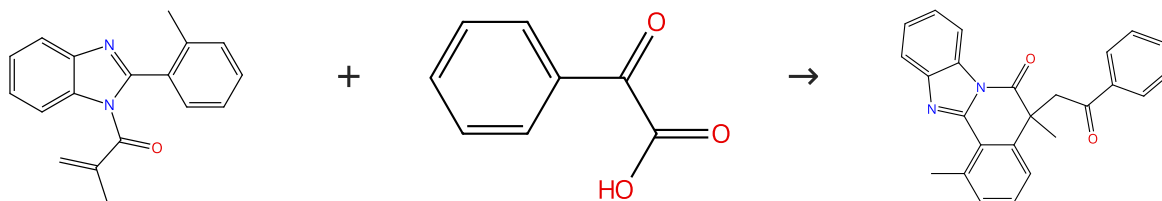
Organic Letters (2020), 22(17), 6832-6837.

1.1 Reagents: Lithium carbonate (Li_2CO_3)Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy- κ O)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine

Solvents: Dimethylformamide, Water; 24 h, cooled

Scheme 24 (1 Reaction)

Steps: 1 Yield: 84%



Suppliers (105)

31-484-CAS-23434180

Steps: 1 Yield: 84%

Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles

By: Zhu, Hu-Lin; et al

Organic Letters (2021), 23(8), 2976-2980.

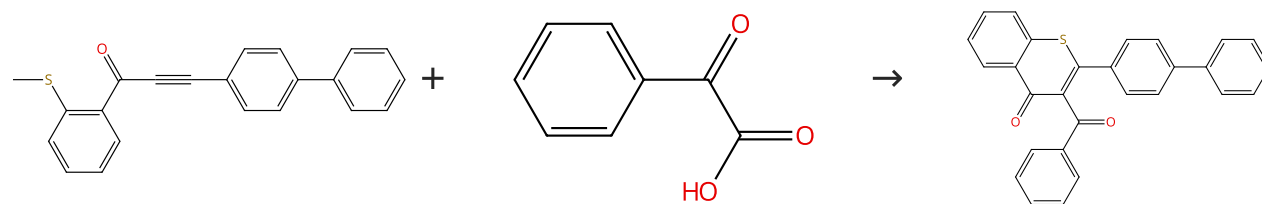
1.1 Reagents: Oxygen

Catalysts: 2,4,5,6-Tetra(9H-carbazol-9-yl)isophthalonitrile

Solvents: Dimethyl carbonate; 6 h, rt

Scheme 25 (1 Reaction)

Steps: 1 Yield: 84%



Suppliers (105)

31-484-CAS-23433946

Steps: 1 Yield: 84%

Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles

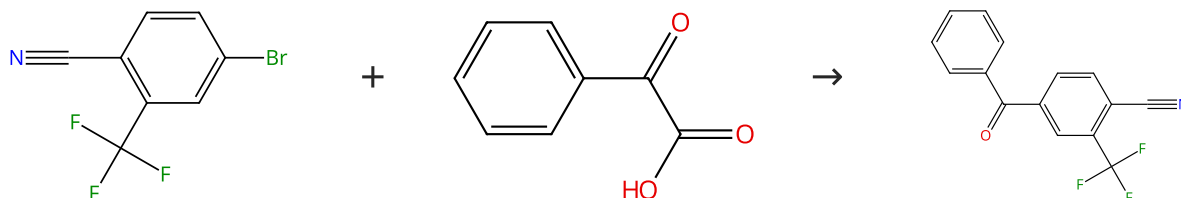
By: Zhu, Hu-Lin; et al

Organic Letters (2021), 23(8), 2976-2980.

1.1 **Reagents:** Benzoyl peroxide
Catalysts: 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile
Solvents: Acetonitrile; 12 h, rt

Scheme 26 (1 Reaction)

Steps: 1 Yield: 84%



Suppliers (85)

Suppliers (105)

31-484-CAS-22486039

Steps: 1 Yield: 84%

Acyl Radicals from α -Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones

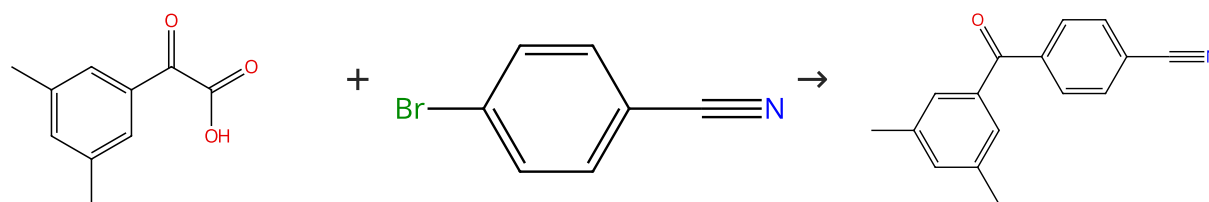
By: Zhu, Da-Liang; et al

Organic Letters (2020), 22(17), 6832-6837.

1.1 **Reagents:** Lithium carbonate (Li_2CO_3)
Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy- κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Dimethylformamide, Water; 24 h, cooled

Scheme 27 (1 Reaction)

Steps: 1 Yield: 84%



Suppliers (19)

Suppliers (92)

31-484-CAS-22486062

Steps: 1 Yield: 84%

Acyl Radicals from α -Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones

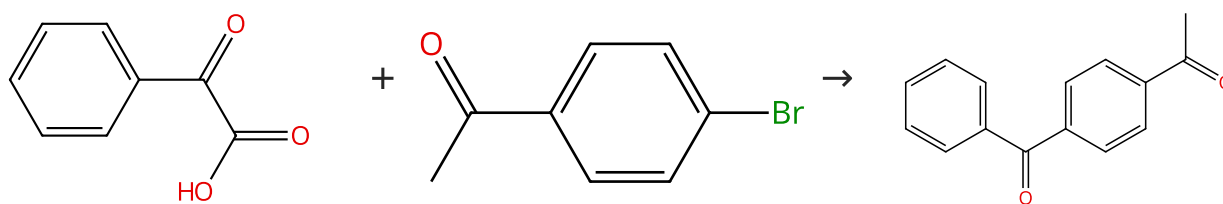
By: Zhu, Da-Liang; et al

Organic Letters (2020), 22(17), 6832-6837.

1.1 **Reagents:** Lithium carbonate (Li_2CO_3)
Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy- κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Dimethylformamide, Water; 24 h, cooled

Scheme 28 (1 Reaction)

Steps: 1 Yield: 84%



Suppliers (105)

Suppliers (84)

Suppliers (22)

31-484-CAS-22486032

Steps: 1 Yield: 84%

Acyl Radicals from α -Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones

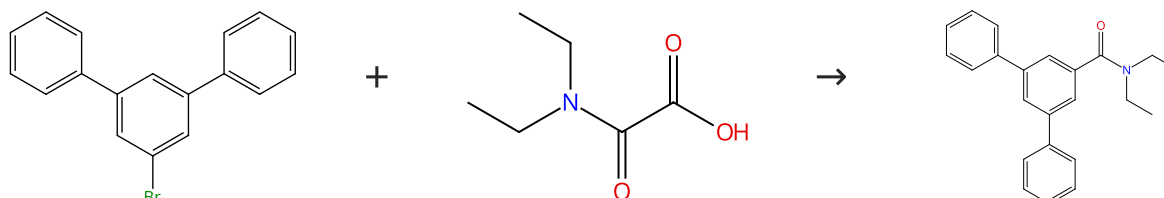
By: Zhu, Da-Liang; et al

Organic Letters (2020), 22(17), 6832-6837.

- 1.1 **Reagents:** Lithium carbonate (Li_2CO_3)
Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy- κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Dimethylformamide, Water; 24 h, cooled

Scheme 29 (1 Reaction)

Steps: 1 Yield: 83%



Suppliers (66)

Suppliers (46)

31-614-CAS-38656515

Steps: 1 Yield: 83%

Photoredox Ni-catalyzed decarboxylative arylation of oxamic acids for synthesis

By: Duan, Depeng; et al

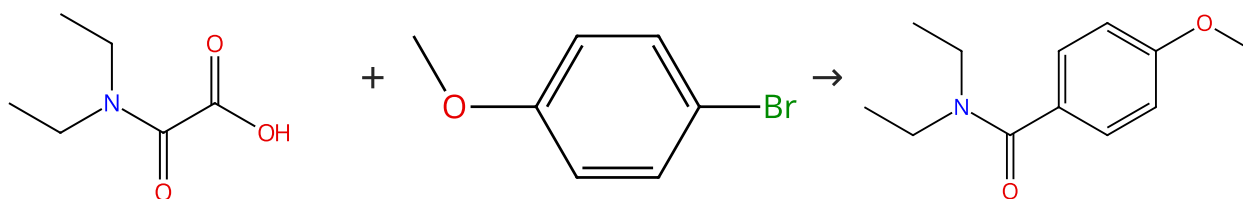
Organic Chemistry Frontiers (2024), 11(1), 47-52.

- 1.1 **Catalysts:** 4,4'-Dimethoxy-2,2'-bipyridine, Dibromo[1,2-di(methoxy- κO)ethane]nickel
Solvents: Acetonitrile; 30 min, rt
- 1.2 **Reagents:** Sodium carbonate
Catalysts: Benzonitrile, 2,3,4,5,6-penta-9*H*-carbazol-9-yl-
Solvents: 1,4-Dioxane; 12 h, rt

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 83%



Suppliers (46)

Suppliers (71)

Suppliers (41)

31-614-CAS-38656519

Steps: 1 Yield: 83%

Photoredox Ni-catalyzed decarboxylative arylation of oxamic acids for synthesis

By: Duan, Depeng; et al

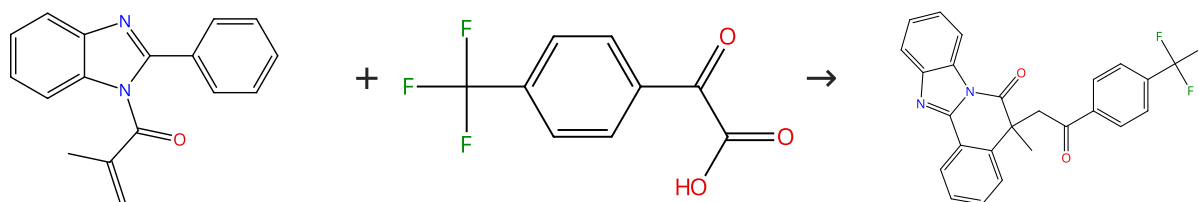
Organic Chemistry Frontiers (2024), 11(1), 47-52.

- 1.1 **Catalysts:** 4,4'-Dimethoxy-2,2'-bipyridine, Dibromo[1,2-di(methoxy- κO)ethane]nickel
Solvents: Acetonitrile; 30 min, rt
- 1.2 **Reagents:** Sodium carbonate
Catalysts: Benzonitrile, 2,3,4,5,6-penta-9*H*-carbazol-9-yl-
Solvents: 1,4-Dioxane; 12 h, rt

Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 83%



Suppliers (45)

31-484-CAS-23433418

Steps: 1 Yield: 83%

Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles

1.1 Reagents: Oxygen

Catalysts: 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile

Solvents: Dimethyl carbonate; 6 h, rt

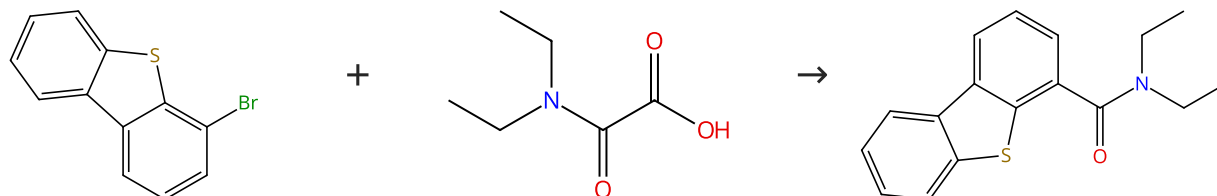
By: Zhu, Hu-Lin; et al

Organic Letters (2021), 23(8), 2976-2980.

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 83%



Suppliers (76)

Suppliers (46)

Suppliers (3)

31-614-CAS-38656525

Steps: 1 Yield: 83%

Photoredox Ni-catalyzed decarboxylative arylation of oxamic acids for synthesis

1.1 Catalysts: 4,4'-Dimethoxy-2,2'-bipyridine, Dibromo[1,2-di(methoxy- κ O)ethane]nickel

Solvents: Acetonitrile; 30 min, rt

By: Duan, Depeng; et al

Organic Chemistry Frontiers (2024), 11(1), 47-52.

1.2 Reagents: Sodium carbonate

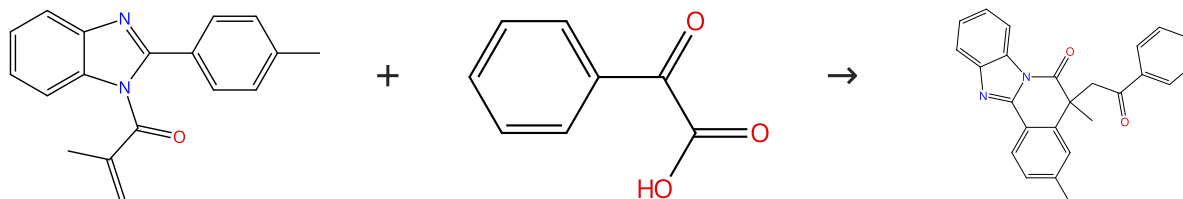
Catalysts: Benzonitrile, 2,3,4,5,6-penta-9*H*-carbazol-9-yl-

Solvents: 1,4-Dioxane; 12 h, rt

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 82%



Suppliers (105)

31-484-CAS-23432527

Steps: 1 Yield: 82%

Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles

1.1 Reagents: Oxygen

Catalysts: 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile

Solvents: Dimethyl carbonate; 6 h, rt

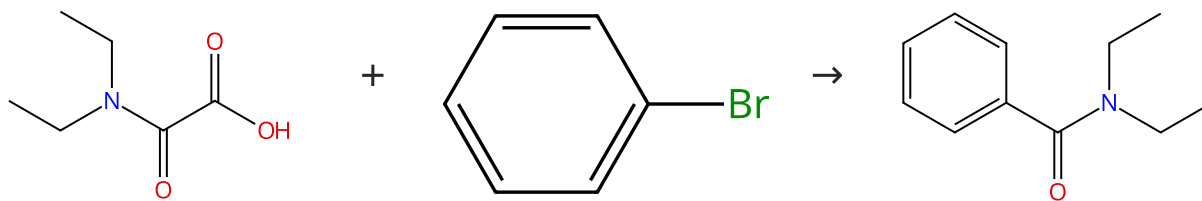
By: Zhu, Hu-Lin; et al

Organic Letters (2021), 23(8), 2976-2980.

Experimental Protocols

Scheme 34 (1 Reaction)

Steps: 1 Yield: 82%



Suppliers (46)

Suppliers (81)

Suppliers (64)

31-614-CAS-38656518

Steps: 1 Yield: 82%

Photoredox Ni-catalyzed decarboxylative arylation of oxamic acids for synthesis

By: Duan, Depeng; et al

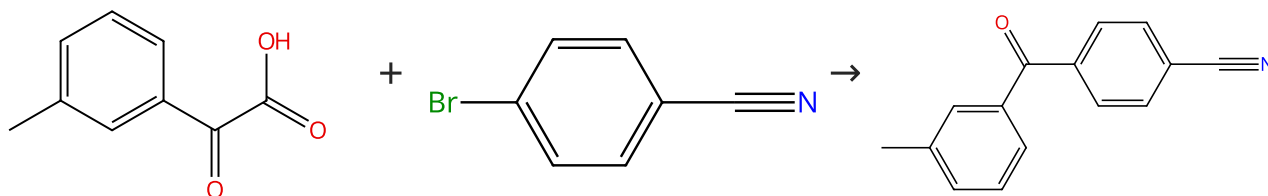
Organic Chemistry Frontiers (2024), 11(1), 47-52.

1.1 **Catalysts:** 4,4'-Dimethoxy-2,2'-bipyridine, Dibromo[1,2-di(methoxy-κO)ethane]nickel**Solvents:** Acetonitrile; 30 min, rt1.2 **Reagents:** Sodium carbonate**Catalysts:** 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile**Solvents:** 1,4-Dioxane; 12 h, rt

Experimental Protocols

Scheme 35 (1 Reaction)

Steps: 1 Yield: 82%



Suppliers (40)

Suppliers (92)

Suppliers (5)

31-484-CAS-22486061

Steps: 1 Yield: 82%

Acyl Radicals from α-Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones

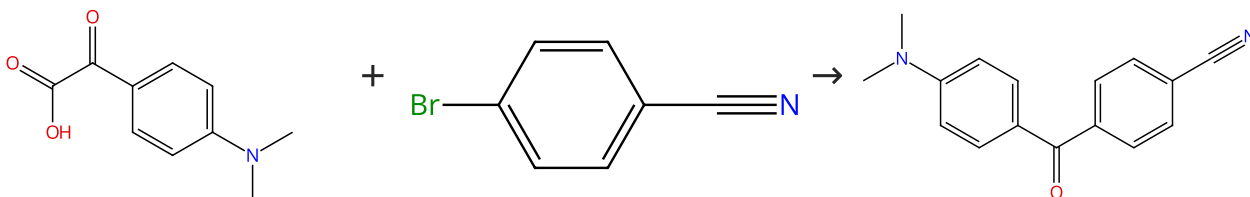
By: Zhu, Da-Liang; et al

Organic Letters (2020), 22(17), 6832-6837.

1.1 **Reagents:** Lithium carbonate (Li₂CO₃)
Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Dimethylformamide, Water; 24 h, cooled

Scheme 36 (1 Reaction)

Steps: 1 Yield: 82%



Suppliers (17)

Suppliers (92)

31-484-CAS-22486059

Steps: 1 Yield: 82%

Acyl Radicals from α-Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones

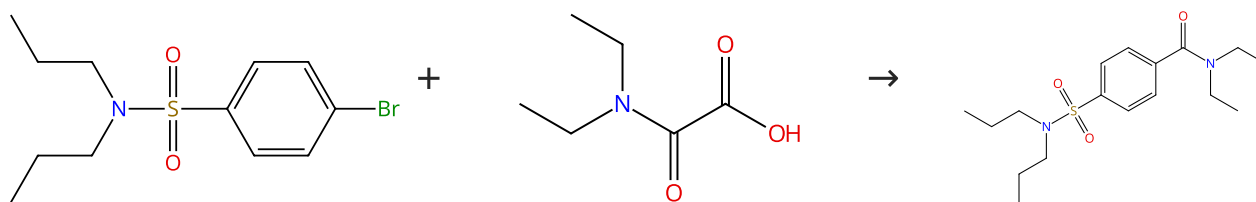
By: Zhu, Da-Liang; et al

Organic Letters (2020), 22(17), 6832-6837.

1.1 **Reagents:** Lithium carbonate (Li₂CO₃)
Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Dimethylformamide, Water; 24 h, cooled

Scheme 37 (1 Reaction)

Steps: 1 Yield: 81%



Suppliers (33)

Suppliers (46)

Supplier (1)

31-614-CAS-38656528

Steps: 1 Yield: 81%

Photoredox Ni-catalyzed decarboxylative arylation of oxamic acids for synthesis

By: Duan, Depeng; et al

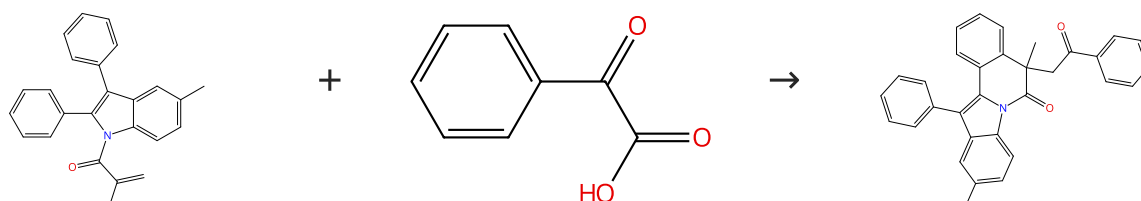
Organic Chemistry Frontiers (2024), 11(1), 47-52.

- 1.1 **Catalysts:** 4,4'-Dimethoxy-2,2'-bipyridine, Dibromo[1,2-di(methoxy-κO)ethane]nickel
Solvents: Acetonitrile; 30 min, rt
- 1.2 **Reagents:** Sodium carbonate
Catalysts: Benzonitrile, 2,3,4,5,6-penta-9*H*-carbazol-9-yl-
Solvents: 1,4-Dioxane; 12 h, rt

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 81%



Suppliers (105)

31-484-CAS-23433578

Steps: 1 Yield: 81%

Acyl Radicals from α-Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles

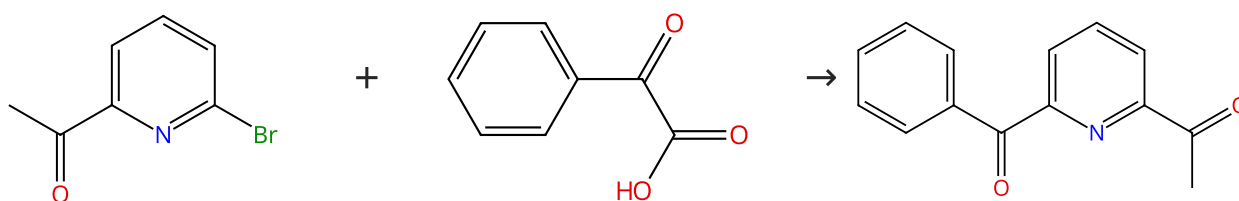
By: Zhu, Hu-Lin; et al

Organic Letters (2021), 23(8), 2976-2980.

- 1.1 **Reagents:** Oxygen
Catalysts: 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile
Solvents: Dimethyl carbonate; 6 h, rt

Scheme 39 (1 Reaction)

Steps: 1 Yield: 80%



Suppliers (87)

Suppliers (105)

31-484-CAS-22486047

Steps: 1 Yield: 80%

Acyl Radicals from α-Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones

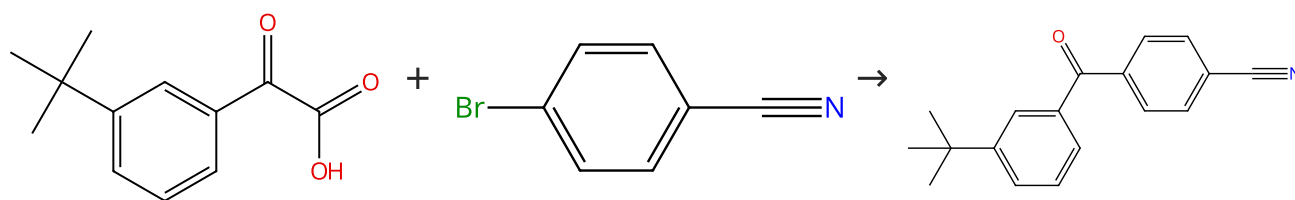
By: Zhu, Da-Liang; et al

Organic Letters (2020), 22(17), 6832-6837.

- 1.1 **Reagents:** Lithium carbonate (Li₂CO₃)
Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Dimethylformamide, Water; 24 h, cooled

Scheme 40 (1 Reaction)

Steps: 1 Yield: 80%



Suppliers (9)

Suppliers (92)

31-484-CAS-23543450

Steps: 1 Yield: 80%

Acyl Radicals from α -Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones

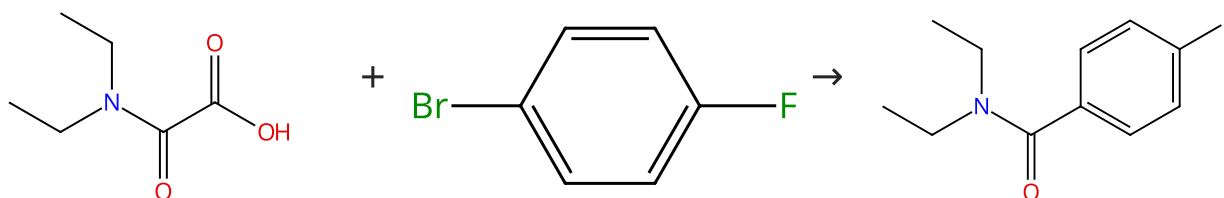
By: Zhu, Da-Liang; et al

Organic Letters (2020), 22(17), 6832-6837.

- 1.1 **Reagents:** Lithium carbonate (Li_2CO_3)
Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy- κ O)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Dimethylformamide, Water; 24 h, cooled

Scheme 41 (1 Reaction)

Steps: 1 Yield: 80%



Suppliers (46)

Suppliers (103)

Suppliers (41)

31-614-CAS-38656510

Steps: 1 Yield: 80%

Photoredox Ni-catalyzed decarboxylative arylation of oxamic acids for synthesis

By: Duan, Depeng; et al

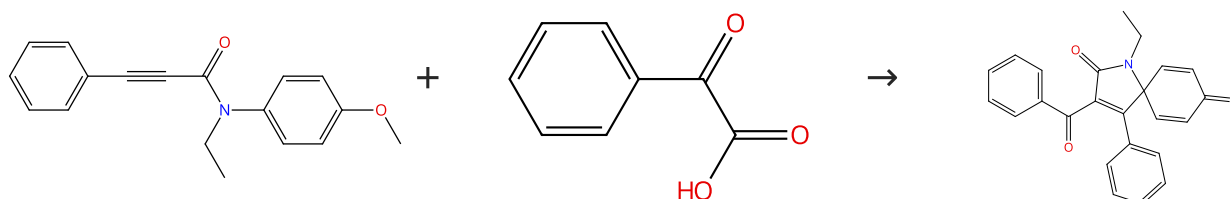
Organic Chemistry Frontiers (2024), 11(1), 47-52.

- 1.1 **Catalysts:** 4,4'-Dimethoxy-2,2'-bipyridine, Dibromo[1,2-di(methoxy- κ O)ethane]nickel
Solvents: Acetonitrile; 30 min, rt
 1.2 **Reagents:** Sodium carbonate
Catalysts: Benzonitrile, 2,3,4,5,6-penta-9H-carbazol-9-yl-
Solvents: 1,4-Dioxane; 12 h, rt

Experimental Protocols

Scheme 42 (1 Reaction)

Steps: 1 Yield: 80%



Suppliers (105)

31-484-CAS-23433687

Steps: 1 Yield: 80%

Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles

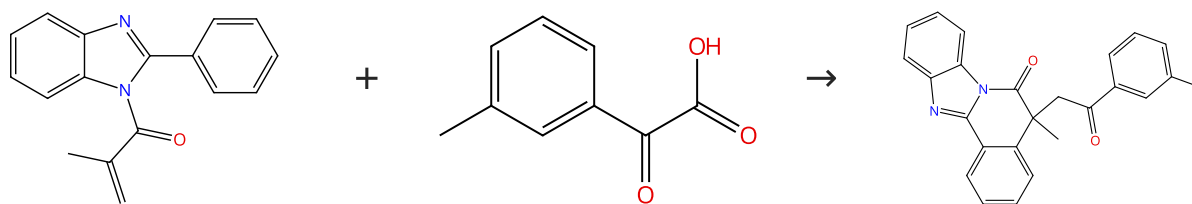
By: Zhu, Hu-Lin; et al

Organic Letters (2021), 23(8), 2976-2980.

- 1.1 **Reagents:** Benzoyl peroxide
Catalysts: 2,4,5,6-Tetra(9H-carbazol-9-yl)isophthalonitrile
Solvents: Acetonitrile; 24 h, rt

Scheme 43 (1 Reaction)

Steps: 1 Yield: 80%



Suppliers (40)

31-484-CAS-23432593

Steps: 1 Yield: 80%

Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles

1.1 Reagents: Oxygen

Catalysts: 2,4,5,6-Tetra(9H-carbazol-9-yl)isophthalonitrile

Solvents: Dimethyl carbonate; 6 h, rt

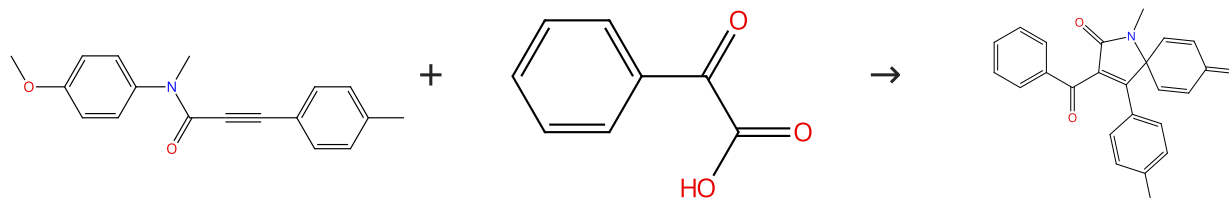
By: Zhu, Hu-Lin; et al

Organic Letters (2021), 23(8), 2976-2980.

Experimental Protocols

Scheme 44 (1 Reaction)

Steps: 1 Yield: 80%



Supplier (1)

Suppliers (105)

31-484-CAS-23432630

Steps: 1 Yield: 80%

Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles

1.1 Reagents: Benzoyl peroxide

Catalysts: 2,4,5,6-Tetra(9H-carbazol-9-yl)isophthalonitrile

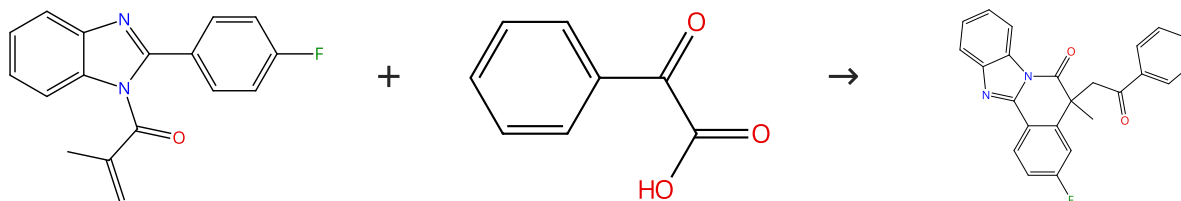
Solvents: Acetonitrile; 24 h, rt

By: Zhu, Hu-Lin; et al

Organic Letters (2021), 23(8), 2976-2980.

Scheme 45 (1 Reaction)

Steps: 1 Yield: 79%



Suppliers (105)

31-484-CAS-23434126

Steps: 1 Yield: 79%

Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles

1.1 Reagents: Oxygen

Catalysts: 2,4,5,6-Tetra(9H-carbazol-9-yl)isophthalonitrile

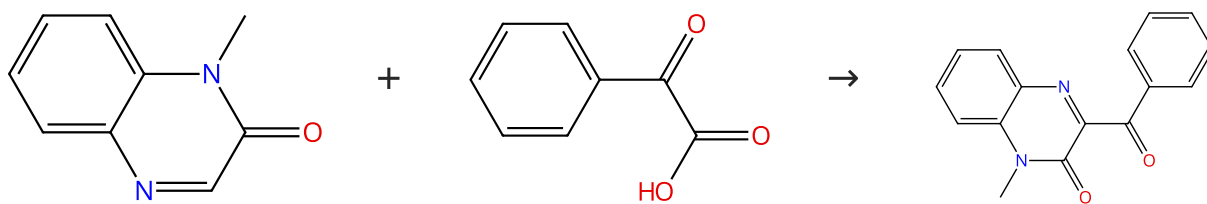
Solvents: Dimethyl carbonate; 6 h, rt

By: Zhu, Hu-Lin; et al

Organic Letters (2021), 23(8), 2976-2980.

Scheme 46 (1 Reaction)

Steps: 1 Yield: 79%



Suppliers (43)

Suppliers (105)

Suppliers (3)

31-219-CAS-23433606

Steps: 1 Yield: 79%

Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles

1.1 Reagents: Oxygen

Catalysts: 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile

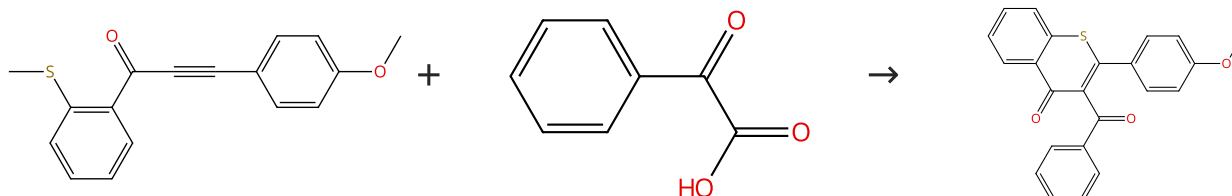
Solvents: Dimethyl carbonate; 6 h, rt

By: Zhu, Hu-Lin; et al

Organic Letters (2021), 23(8), 2976-2980.

Scheme 47 (1 Reaction)

Steps: 1 Yield: 78%



Suppliers (105)

31-484-CAS-23432814

Steps: 1 Yield: 78%

Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles

1.1 Reagents: Benzoyl peroxide

Catalysts: 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile

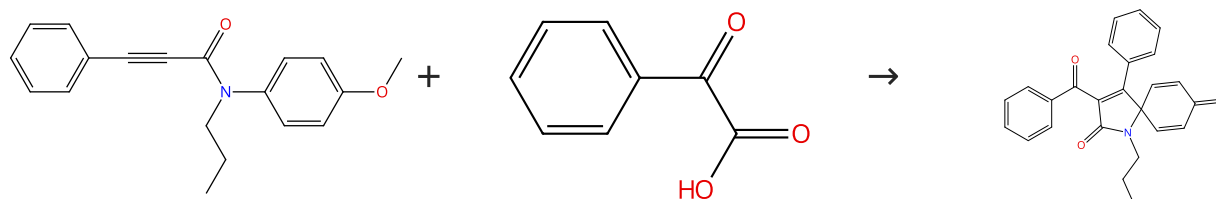
Solvents: Acetonitrile; 12 h, rt

By: Zhu, Hu-Lin; et al

Organic Letters (2021), 23(8), 2976-2980.

Scheme 48 (1 Reaction)

Steps: 1 Yield: 78%



Suppliers (105)

31-484-CAS-23432404

Steps: 1 Yield: 78%

Acyl Radicals from α -Keto Acids: Metal-Free Visible-Light-Promoted Acylation of Heterocycles

1.1 Reagents: Benzoyl peroxide

Catalysts: 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile

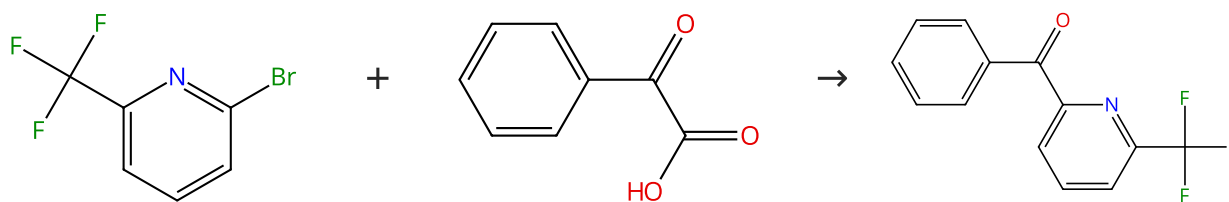
Solvents: Acetonitrile; 24 h, rt

By: Zhu, Hu-Lin; et al

Organic Letters (2021), 23(8), 2976-2980.

Scheme 49 (1 Reaction)

Steps: 1 Yield: 78%



Suppliers (85)

Suppliers (105)

31-484-CAS-22486045

Steps: 1 Yield: 78%

Acyl Radicals from α -Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones

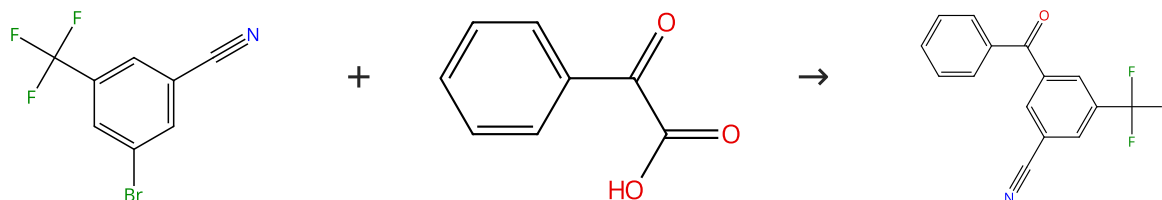
1.1 **Reagents:** Lithium carbonate (Li_2CO_3)
Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy- κ O)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Dimethylformamide, Water; 24 h, cooled

By: Zhu, Da-Liang; et al

Organic Letters (2020), 22(17), 6832-6837.

Scheme 50 (1 Reaction)

Steps: 1 Yield: 78%



Suppliers (67)

Suppliers (105)

31-484-CAS-22486041

Steps: 1 Yield: 78%

Acyl Radicals from α -Keto Acids Using a Carbonyl Photocatalyst: Photoredox-Catalyzed Synthesis of Ketones

1.1 **Reagents:** Lithium carbonate (Li_2CO_3)
Catalysts: 2-Chlorothioxanthone, Dichloro[1,2-di(methoxy- κ O)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine
Solvents: Dimethylformamide, Water; 24 h, cooled

By: Zhu, Da-Liang; et al

Organic Letters (2020), 22(17), 6832-6837.